

LOAN RISK & AUDIT ANALYSIS USING SQL

Portfolio Project

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Role: Data Analyst

Tools Used: SQL (JOINS, Common Table Expressions (CTEs), CASE Statements, Aggregate Functions, GROUP BY, ORDER BY), Microsoft Excel.

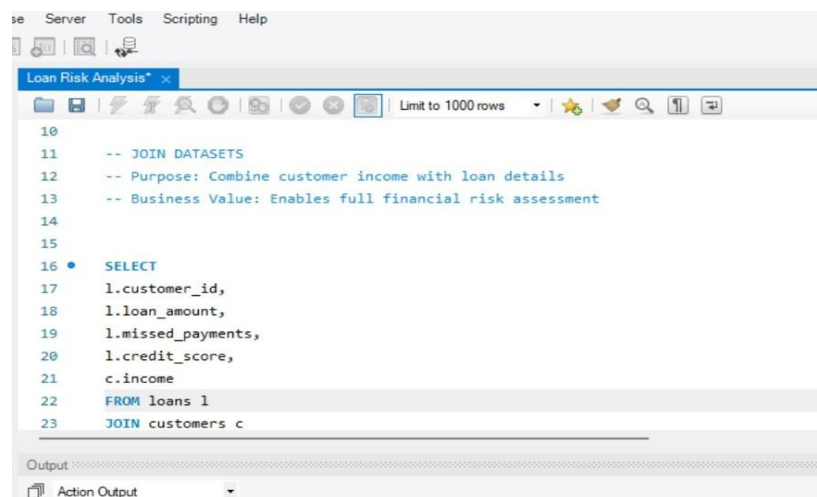
Executive Summary

The objective of this project was to identify high-risk loan customers by analysing customer and loan datasets using SQL. The analysis combined customer financial information with loan performance data to identify borrowers who present a higher probability of default.

Using SQL techniques such as INNER JOINS, Common Table Expressions (CTEs), CASE statements, aggregate functions and calculated financial ratios, a reusable analytical dataset was developed to classify customers into High Risk, Medium Risk and Low Risk categories.

From an accounting and audit perspective, the analysis demonstrates how deteriorating credit quality can affect a bank's loan portfolio, impairment provisions, profitability and financial position.

This project demonstrates practical SQL querying skills, business analysis, financial risk assessment and accounting interpretation.



1. Project Objective

The primary objective of this project was to use SQL to analyse banking data and identify customers who may default on their loan repayments.

The project focused on:

- *Combining customer and loan datasets
- *Creating reusable SQL queries using CTEs
- *Identifying customers with high default risk
- *Measuring financial burden using Loan-to-Income Ratio
- *Producing business insights for lending decisions
- *Explaining the accounting impact of loan defaults

2. Dataset Overview

The project uses two related datasets.

Bank Customers Dataset

- *Contains customer information including:
- *Customer ID
- *Customer Income
- *Customer Demographic Information

Bank Loan Dataset

- *Contains loan information including:
- *Customer ID
- *Loan Amount
- *Credit Score
- *Missed Payments

The two datasets were linked using the Customer_ID field.

3. SQL Methodology

The project followed a structured SQL workflow.

Step 1 – Preview the Data

The initial SQL queries were used to understand both datasets.

```
SELECT * FROM loans;
```

```
SELECT * FROM customers;
```

This ensured both tables were correctly loaded before analysis.

Step 2 – Join Both Datasets

An INNER JOIN combined customer income with loan information.

```
SELECT  
l.customer_id,  
l.loan_amount,  
l.missed_payments,  
l.credit_score,  
c.income  
FROM loans l  
JOIN customers c  
ON l.customer_id = c.customer_id;
```

Business Value

Combining both datasets provides a complete customer financial profile.

Without the JOIN, meaningful risk analysis would not be possible.

Step 3 – Create a Common Table Expression (CTE)

A Common Table Expression (CTE) was created to make the SQL script cleaner and easier to reuse.

```
WITH loan_data AS (  
SELECT ...  
)  
SELECT *  
FROM loan_data;
```

Why CTE?

The CTE creates a reusable dataset that can be referenced multiple times without repeating the JOIN logic.

This improves:

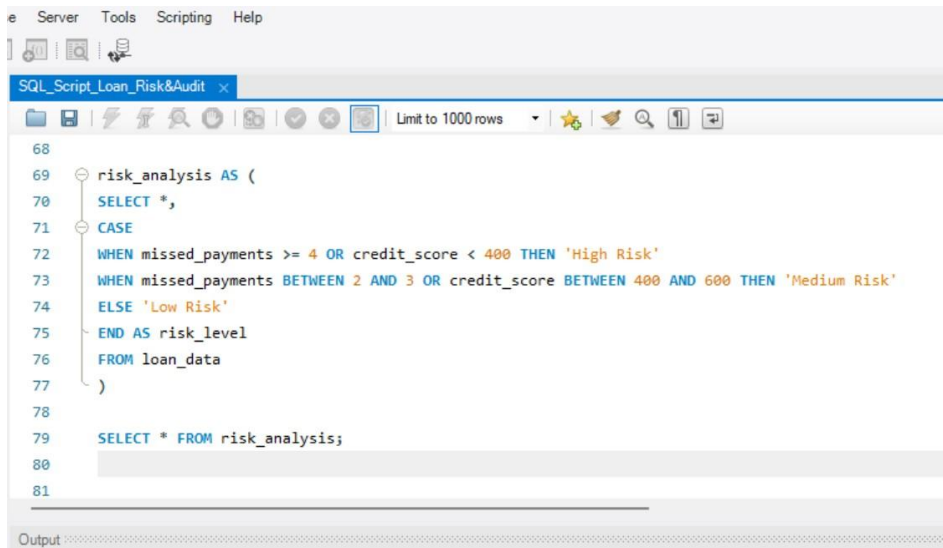
Readability

Maintainability

Query performance

Scalability

Step 4 – Risk Classification

A screenshot of a SQL IDE window titled 'SQL_Script_Loan_Risk&Audit'. The window has a menu bar with 'e', 'Server', 'Tools', 'Scripting', and 'Help'. Below the menu bar is a toolbar with various icons. The main area displays a SQL script with line numbers 68 to 81. The script defines a table function 'risk_analysis' and then queries it. The script is as follows:

```
68  
69 risk_analysis AS (  
70   SELECT *,  
71   CASE  
72     WHEN missed_payments >= 4 OR credit_score < 400 THEN 'High Risk'  
73     WHEN missed_payments BETWEEN 2 AND 3 OR credit_score BETWEEN 400 AND 600 THEN 'Medium Risk'  
74     ELSE 'Low Risk'  
75   END AS risk_level  
76   FROM loan_data  
77 )  
78  
79 SELECT * FROM risk_analysis;  
80  
81
```

Customers were classified into three risk categories.

High Risk

Customers with:

Missed Payments ≥ 4

OR

Credit Score < 400

Medium Risk

Customers with:

Missed Payments between 2 and 3

OR

Credit Score between 400 and 600

Low Risk

Customers with:

Missed Payments < 2

AND

Credit Score > 600

This classification was implemented using SQL CASE statements.

Step 5 – Loan-to-Income Ratio

Another SQL query calculated:

$\text{Loan Amount} \div \text{Customer Income}$

This metric measures the customer's repayment capacity.

Customers with higher ratios are more financially stretched and therefore more likely to default.

The results were ordered in descending order to prioritise the highest-risk customers.

4. Key Findings

1. Credit Score is the strongest predictor of risk

Customers with very low credit scores consistently fall into the High Risk category.

A poor credit score reflects previous borrowing behaviour and indicates a greater likelihood of default.

2. Missed Payments increase default risk

Borrowers with four or more missed repayments are significantly more likely to become non-performing loans.

Payment history is therefore one of the most reliable indicators of future repayment behaviour.

3. High Loan-to-Income Ratio signals financial strain

Customers with larger loans relative to their income have less financial flexibility.

These customers are more vulnerable to:

- *Unemployment
- *Inflation
- *Interest rate increases
- *Unexpected expenses

4. Combining multiple indicators improves risk assessment

Rather than relying on one variable alone, combining:

- *Credit Score
- *Missed Payments
- *Income
- *Loan Amount

provides a much more reliable assessment of borrower risk.

5. Accounting Perspective

From an accounting standpoint, loan defaults directly affect a bank's financial statements.

Expected Credit Losses (ECL)

High-risk loans require higher Expected Credit Loss provisions.

These provisions reduce reported profit.

Balance Sheet Impact

When loans become impaired:

- *Loan Assets decrease
- *Allowance for Credit Losses increases
- *Net Loan Value decreases

This weakens the bank's overall financial position.

Income Statement Impact

Loan defaults reduce:

- *Interest Income

while increasing:

- *Bad Debt Expense
- *Impairment Expense

As a result:

- *Net Profit decreases.

Cash Flow Impact

When borrowers stop repaying loans:

- *Cash inflows decrease

while

- *Collection costs increase.

This negatively affects liquidity.

6. Audit Perspective

This analysis also supports internal audit activities.

Auditors can use the results to:

- *Identify risky borrowers
- *Review loan approval controls
- *Evaluate impairment calculations
- *Assess compliance with lending policies
- *Improve monitoring of overdue accounts
- *Strengthen internal control systems

7. Business Recommendations

Based on the SQL analysis, the following recommendations are proposed:

Improve Credit Approval Policies

Use minimum credit score thresholds before approving loans.

Monitor High Loan-to-Income Customers

Customers with excessive debt relative to income should receive enhanced monitoring.

Introduce Early Warning Systems

Automatically flag customers who:

- *miss repayments
- *experience declining credit scores
- *exceed acceptable debt ratios

Increase Portfolio Monitoring

Regularly review high-risk borrowers before they become non-performing loans.

Expand Predictive Analytics

Future work could incorporate machine learning models to predict defaults before they occur.

8. Technical Skills Demonstrated

This project demonstrates proficiency in:

- *SQL
- *INNER JOIN
- *Common Table Expressions (CTEs)
- *CASE Statements
- *Aggregate Functions

- *GROUP BY
- *ORDER BY
- *Financial Ratio Analysis
- *Risk Classification
- *Credit Risk Analytics
- *Business Intelligence
- *Accounting Interpretation
- *Audit Analytics
- *Data Storytelling

9. Conclusion

This project successfully demonstrates the application of SQL to solve a real-world banking problem by identifying high-risk borrowers through structured data analysis. By integrating customer and loan datasets, classifying borrowers using financial indicators, and calculating loan-to-income ratios, the analysis provides actionable insights that support better lending decisions and stronger risk management.

From an accounting perspective, the findings highlight how deteriorating loan quality can lead to higher impairment provisions, reduced profitability, lower asset values, and increased expected credit losses. From an audit perspective, the analysis strengthens internal controls by enabling early identification of customers with elevated default risk.

Overall, this project showcases my ability to combine SQL programming, data analytics, accounting knowledge, and business problem-solving to deliver practical insights that improve decision-making in the banking and financial services sector.

10. Key Competencies Demonstrated Include:

- *SQL
- *Data Cleaning & Transformation
- *Data Analysis
- *Risk Analytics
- *Financial Analysis
- *Banking & Credit Risk
- *Accounting Knowledge
- *Business Intelligence

*Problem Solving

*Analytical Reporting

*Attention to Detail

*Stakeholder Communication